

The Universe & Our Solar System

This is your guided version. It uses shorter sentences, word banks, and fill-in-the-blank notes to help you learn one step at a time. Read slowly. Fill in every blank. You've got this!



Georgia Standard S6E1 • Supported Reading

WHAT YOU'LL LEARN

01 How Our View Changed

03 Meet the Planets

05 Gravity & Inertia

02 Our Place in the Universe

04 Comparing the Planets

06 Comets, Asteroids & Meteoroids



Name: _____

Period: _____

Date: _____

1.1 How Our View of Space Changed

My Learning Goal

I can explain how scientists' model of space **changed** when they got new evidence.

A **model** is a simple way to show something hard to see.

Long ago, people made models of space using only their eyes.

Later, they built tools like the **telescope**. The telescope gave them new evidence.

When they got new evidence, they **changed** their model. This is how science works.

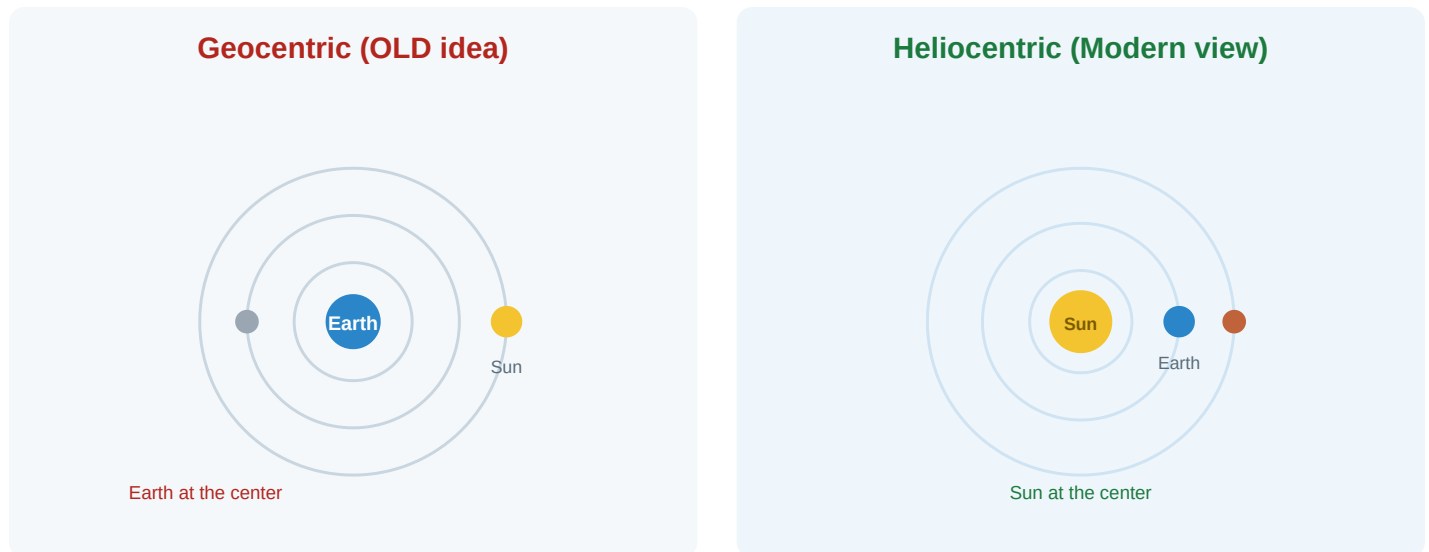


Figure 1.1 — The old model put Earth in the middle. The new model puts the Sun in the middle.

Word Bank (use these to fill the blanks below)

Sun

Earth

telescope

evidence

geocentric

heliocentric

Fill in each blank using the word bank:

- The **geocentric** model says _____ is in the center. (Hint: "geo" = Earth)
- The **heliocentric** model says the _____ is in the center. (Hint: "helio" = Sun)
- The tool that gave scientists new evidence was the _____.
- Scientists change their model when they find new _____.

Remember This

Science ideas are the **best answer we have right now**. New evidence can change them. That is a good thing!

1.2 Our Place in the Universe

My Learning Goal

I can list the levels of space from smallest (Earth) to largest (the universe).

Earth is small compared to space. Scientists "zoom out" to see where we are.

Where Are We? Zooming Out



Figure 1.2 — Zooming out from Earth to the whole universe. Each step is much bigger.

Put the levels in order. Write **1** for smallest and **4** for largest:

Level	What It Is	Order (1–4)
The Universe	Everything that exists	_____
Earth	Our planet	_____
Milky Way Galaxy	Billions of stars, including our Sun	_____
Solar System	The Sun and everything orbiting it	_____

Finish the sentence: Our solar system is part of the _____ galaxy.

1.3 Meet the Planets

My Learning Goal

I can name the 8 planets in order from the Sun.

Eight planets orbit the Sun. Use this sentence to remember the order:

My Very Excellent Mother Just Served Us Nachos

Order	First Letter	Planet Name
1	M	_____
2	V	_____
3	E	_____
4	M	_____
5	J	_____
6	S	_____
7	U	_____
8	N	_____

The first 4 planets are small and **rocky**. The last 4 are huge **gas and ice giants**.

Watch Out

Pluto is **not** a planet anymore. Scientists now call it a *dwarf planet*. The model changed!

1.4 Comparing the Planets

My Learning Goal

I can explain why Earth can support life but other planets cannot.

Earth is the only planet we know that has life. Here is why:

- Earth has **liquid water**.
- Earth has **air we can breathe**.
- Earth is the **right temperature** — not too hot, not too cold.

Word Bank

too hot

too cold

liquid water

habitable zone

air

Fill in the blanks:

- Earth sits in the _____ — the "just right" distance from the Sun.
- Living things need _____ to survive.
- Venus cannot support life because it is _____.
- Neptune cannot support life because it is _____.

1.5 Why Planets Orbit: Gravity & Inertia

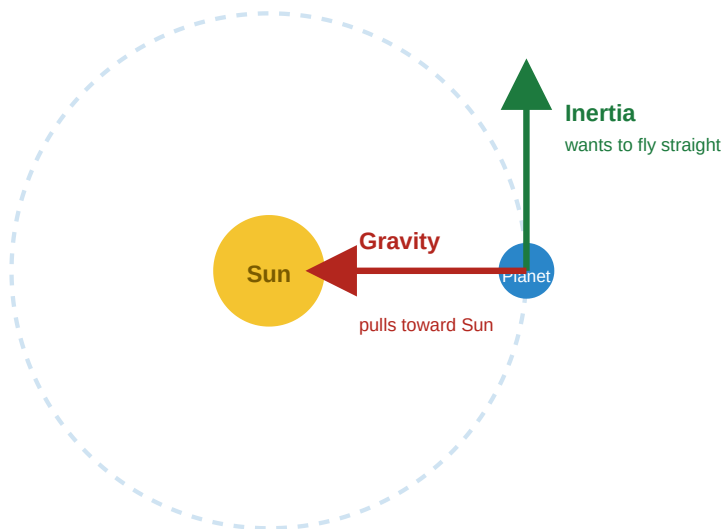
My Learning Goal

I can explain how gravity and inertia work together to make an orbit.

Two forces keep a planet in its orbit:

- **Gravity** = a pull. The Sun's gravity **pulls** the planet toward it.
- **Inertia** = a moving object wants to keep going **straight**.

When these two work together, the planet moves in a **curved path** called an **orbit**.



The two forces combine to make a curved orbit instead of a straight line or a crash.

Figure 1.3 — Gravity pulls in. Inertia goes straight. Together they make a curved orbit.

Easy Way to Picture It

Think of swinging a ball on a string. Your hand pulling = **gravity**. The ball wanting to fly away = **inertia**. The ball keeps going in a circle. That circle is like an **orbit**.

Finish the sentences:

- Gravity _____ the planet toward the Sun. (pulls / pushes)
- Inertia makes the planet want to move in a _____ line. (curved / straight)

1.6 Comets, Asteroids & Meteoroids

My Learning Goal

I can tell the difference between a comet, an asteroid, and a meteoroid.

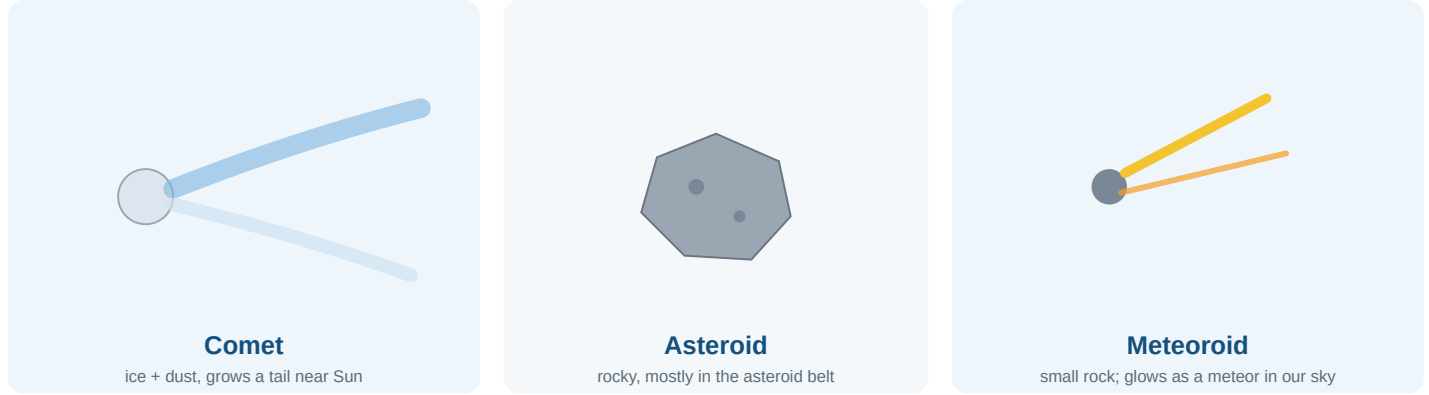


Figure 1.4 — Three kinds of small space objects.

Match each object to what it is made of. Draw a line or write the letter:

Object	Made Of
Comet	A. Rock and metal, in the asteroid belt
Asteroid	B. A small rock that can burn up in our sky
Meteoroid	C. Ice and dust; grows a tail near the Sun

! 3 Names, 1 Rock

- **Meteoroid** = the rock in space.
- **Meteor** = the glowing streak in the sky ("shooting star").
- **Meteorite** = a piece that lands on the ground. (Memory trick: "-ite" **hit** the ground.)

Chapter 1 Review

? Show What You Know

1. The old model with Earth in the center is called the _____ model.
2. Order from smallest to largest: galaxy, Earth, universe, solar system.
3. Name the only planet known to have life: _____.
4. The two forces that keep a planet in orbit are _____ and _____.
5. A small space rock that burns up in our sky is called a _____.